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LayerAlterator: A Raster Processing Tool for Urban Change Simulation

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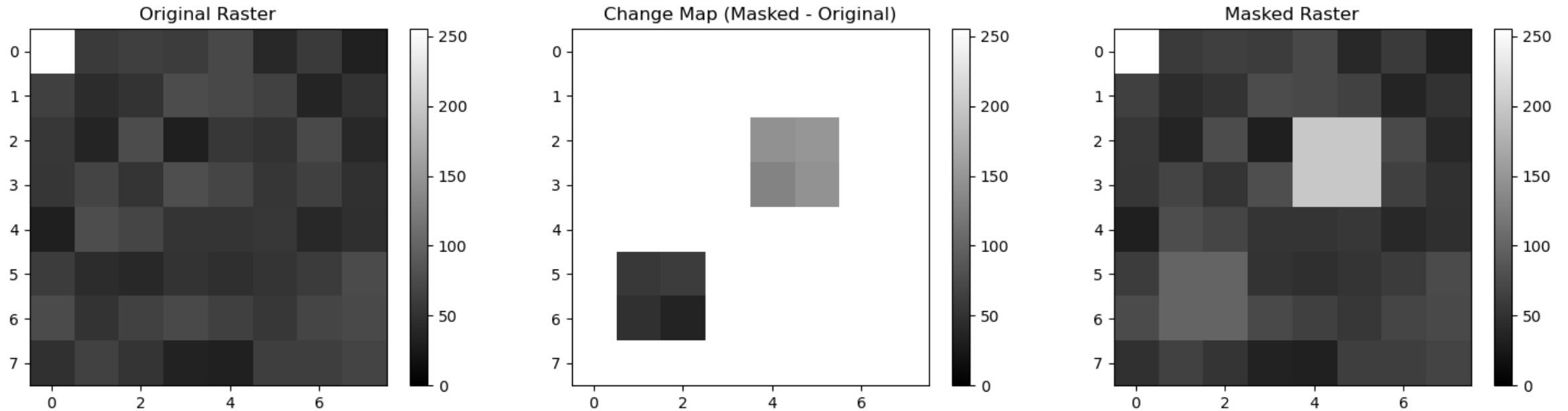
Introduction

Urban Change Modeling: A Need for Lightweight Tools

- Urban planners need to test interventions like reforestation or densification.
- LayerAlterator provides a way to modify such layers through rule-based processing.”

LayerAlterator offers a modular, fast alternative:

- Modify raster layers using **polygon-based rules**
- Supports **masking** (absolute) and **percentage** (relative) updates
- Built with open-source Python libraries



Related Work

Urban Raster Tools & Their Roles

Local Climate Zones (LCZ)

- Stewart & Oke (2012): Classification framework for urban morphology and thermal analysis
- *Useful for urban climate modeling, but static and not designed for layer transformation*

Urban Land Cover Mapping

- Okujeni et al. (2016): Fractional land cover via spectral unmixing
- *Provides accurate input layers for modeling, but lacks editing or simulation logic*

LayerAlterator

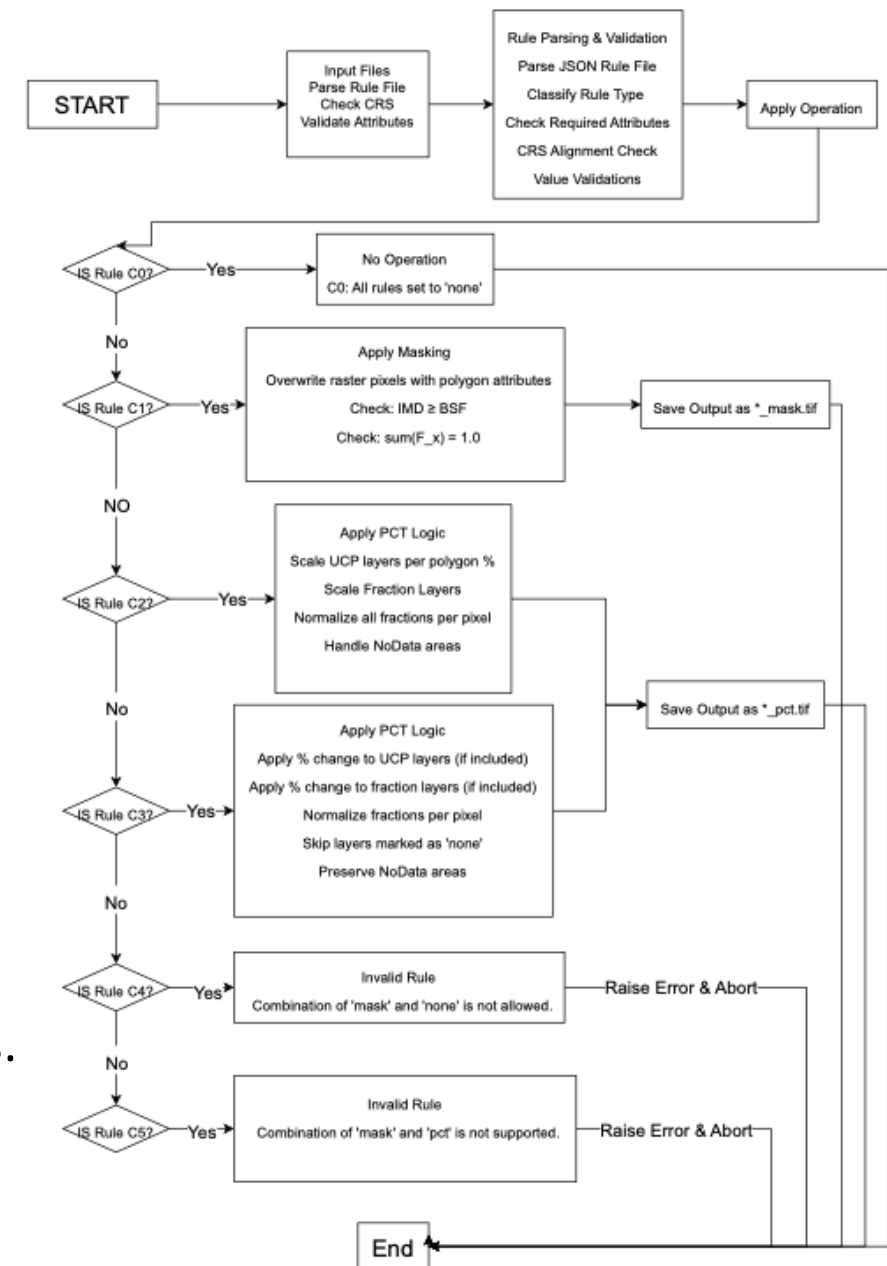
- Designed to modify such input layers using polygon-based rules
- *Prepares customized raster data for use in downstream simulations (“what-if” scenarios)*

Tool	Role	Limitation
LCZ	Urban form classification	Static categories, no editing
Okujeni	High-resolution land cover input	No dynamic modification
LayerAlterator	Prepares input layers for simulations	Not a simulator, but a processing module

Innovation

What Makes LayerAlterator Unique?

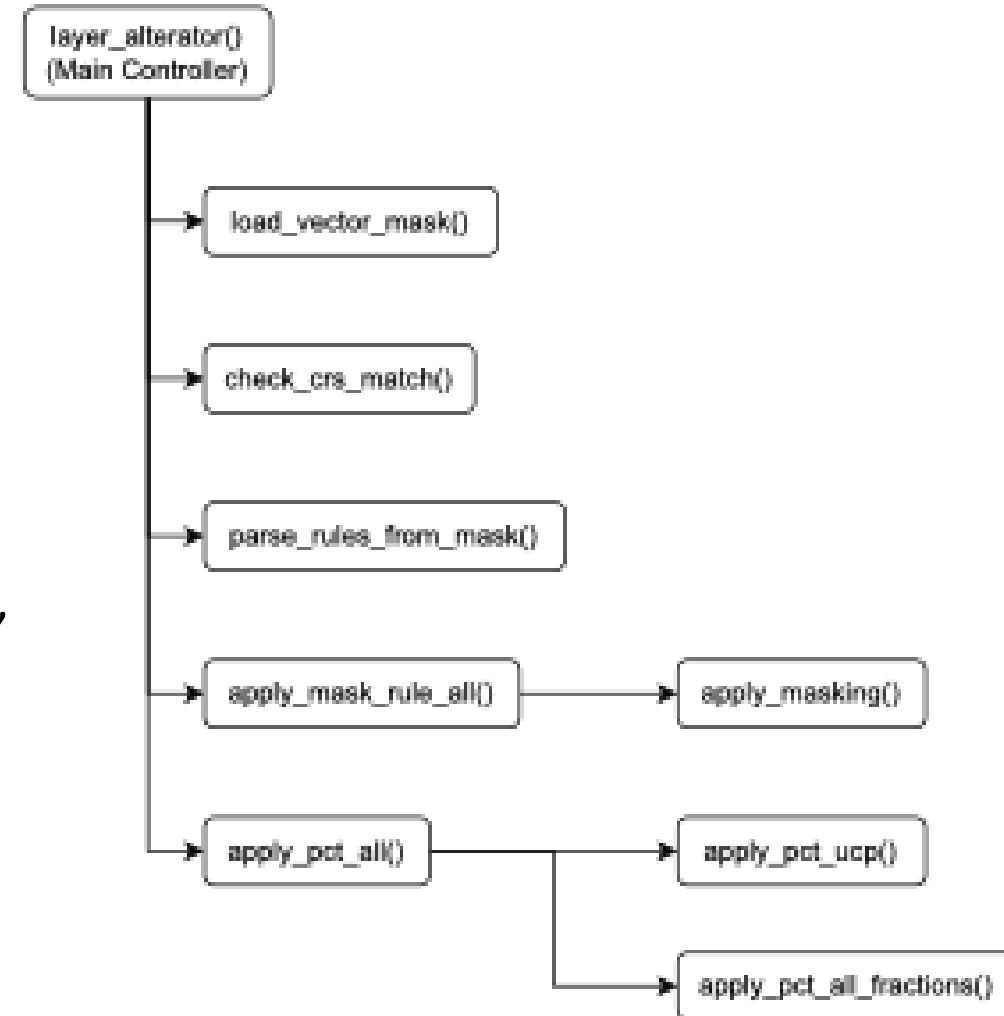
- **Supports both masking and percentage operations**
Apply absolute values *or* modify relative to the current raster data.
- **Rule-driven engine**
Uses simple JSON configuration for flexible simulations.
- **Modular Python architecture**
Easy to extend, debug, or integrate into Digital Twin platforms.
- **Built-in validation**
CRS, normalization, and attribute checks prevent faulty simulations.
- **Transparent and reproducible**
Designed for Jupyter workflows, with visual inspection support.



Architecture Overview

Modular & Reproducible Processing Pipeline

- **Main Controller:** `layer_alterator()` orchestrates the workflow
- **Key Phases:**
 - ❑ Input Loading: `load_vector_mask()`, `check_crs_match()`
 - ❑ Rule Handling: `parse_rules_from_mask()`,
 - ❑ Masking Ops: `apply_masking()`, `apply_mask_rule_all()`,
 - ❑ Percentage Ops: `apply_pct_ucp()`, `apply_pct_all_fractions()`, `apply_pct_all()`
- Functions are independently testable and integrated in Jupyter
- Validation embedded in every major step



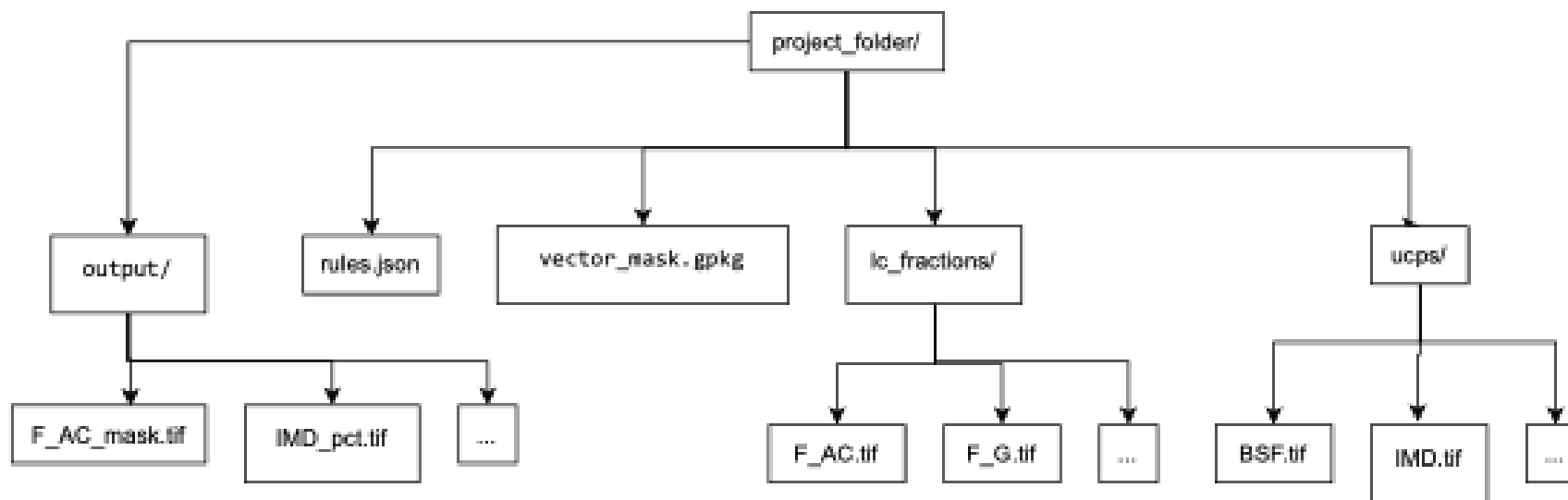
Input and Rule Logic

Input Types

- **Vector Mask:** Polygons (e.g., .gpkg, .geojson) with attribute values
- **Raster Layers:**
- **Rule File:** rules.json defines layer operations

Rule Logic

- Operations: mask, pct, none
- Classified into rule types (C0 to C5)
- Invalid mixes (e.g., mask + pct) trigger errors



Output Example

Traceable Raster Output

- **Processed rasters saved in the output/ folder**
- **Naming Convention:**
 - ❑ *_mask.tif → layers updated with fixed values (masking)
 - ❑ *_pct.tif → layers updated with percentage changes
- **CRS, NoData, and resolution preserved**
- **Visual verification in QGIS (e.g., coverage, alignment, NoData integrity)**
- **Metadata and filenames support scenario comparison**

Name	Name
 BH.tif	 BH_mask.tif
 BH.tif.aux.xml	 BH_mask.tif.aux.xml
 BSF.tif	 BSF_mask.tif
 BSF.tif.aux.xml	 BSF_mask.tif.aux.xml
 IMD.tif	 F_AC_mask.tif
 SVF.tif	 F_AC_mask.tif.aux.xml
 SVF.tif.aux.xml	 F_BS_mask.tif
 TCH.tif	 F_BS_mask.tif.aux.xml
 F_AC.tif	 F_G_mask.tif
 F_AC.tif.aux.xml	 F_G_mask.tif.aux.xml
 F_BS.tif	 F_M_mask.tif
 F_BS.tif.aux.xml	 F_M_mask.tif.aux.xml
 F_G.tif	 F_S_mask.tif
 F_G.tif.aux.xml	 F_S_mask.tif.aux.xml
 F_M.tif	 F_TV_mask.tif
 F_M.tif.aux.xml	 F_TV_mask.tif.aux.xml
 F_S.tif	 F_W_mask.tif
 F_S.tif.aux.xml	 F_W_mask.tif.aux.xml
 F_TV.tif	 IMD_mask.tif
 F_TV.tif.aux.xml	 IMD_mask.tif.aux.xml
 F_W.tif	 SVF_mask.tif
 F_W.tif.aux.xml	 SVF_mask.tif.aux.xml
	 TCH_mask.tif
	 TCH_mask.tif.aux.xml

Validation and Testing

Ensuring Data Integrity and Simulation Accuracy

- **Validation Checks (Pre-Processing)**

- ☐ CRS match between vector and rasters
- ☐ Required attributes present
- ☐ Rule logic consistency

- **During Processing**

- ☐ NoData regions preserved
- ☐ Normalization of fractional rasters
- ☐ Warnings for overlapping polygons or abnormal updates

- **Testing Strategy**

- ☐ Unit tests in Jupyter Notebook (test_functions.ipynb)
- ☐ Visual inspection in QGIS
- ☐ Controlled edge-case scenarios

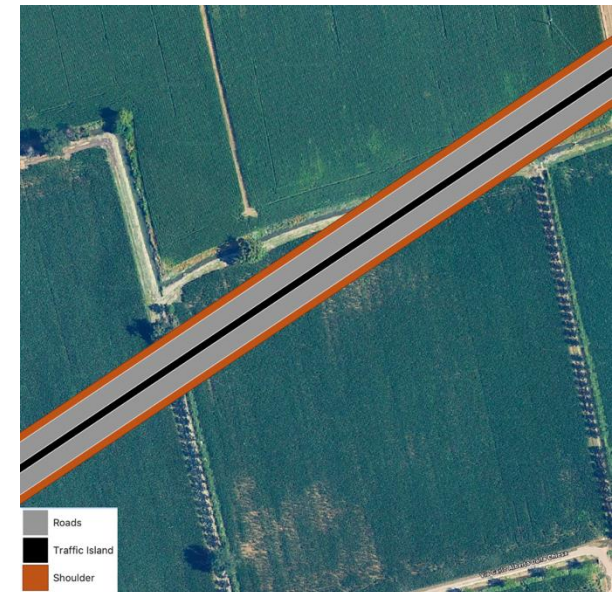
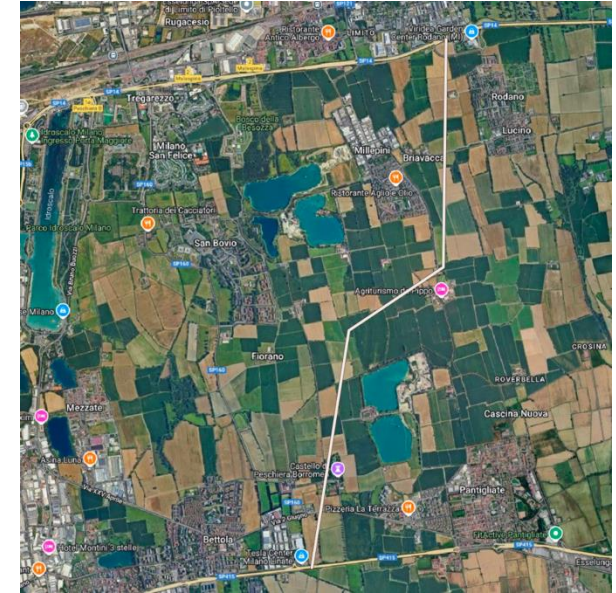
```
--- Test 1: All CRS Match ---  
All raster layers have the same CRS as the vector mask.  
  
--- Test 2: All Rasters Match CRS but Different from Vector ---  
Warning: CRS mismatch detected in the following layers:  
- BH.tif: CRS = EPSG:32632, differs from vector CRS = EPSG:7415  
- F_AC.tif: CRS = EPSG:32632, differs from vector CRS = EPSG:7415  
  
--- Test 3: CRS Mismatch ---  
Warning: Could not open raster for layer IMD_differentCRS.tif: test_data/fun_2/  
IMD_differentCRS.tif: No such file or directory  
All raster layers have the same CRS as the vector mask.  
  
--- Test 4: Missing Raster File ---  
Warning: Could not open raster for layer nonexistent.tif: test_data/fun_2/nonex:  
No such file or directory  
All raster layers have the same CRS as the vector mask.
```

```
--- Test 4: Mix of pct and none ---  
Rule C3: Layers set to PCT and NONE. Proceeding wi  
  
--- Test 5: Mix of mask and none ---  
Rule C4 violation: MASKING + NONE is not allowed.  
  
--- Test 6: Mix of mask and pct ---  
Rule C5 violation: MASKING + PCT is not allowed.
```

Demonstration

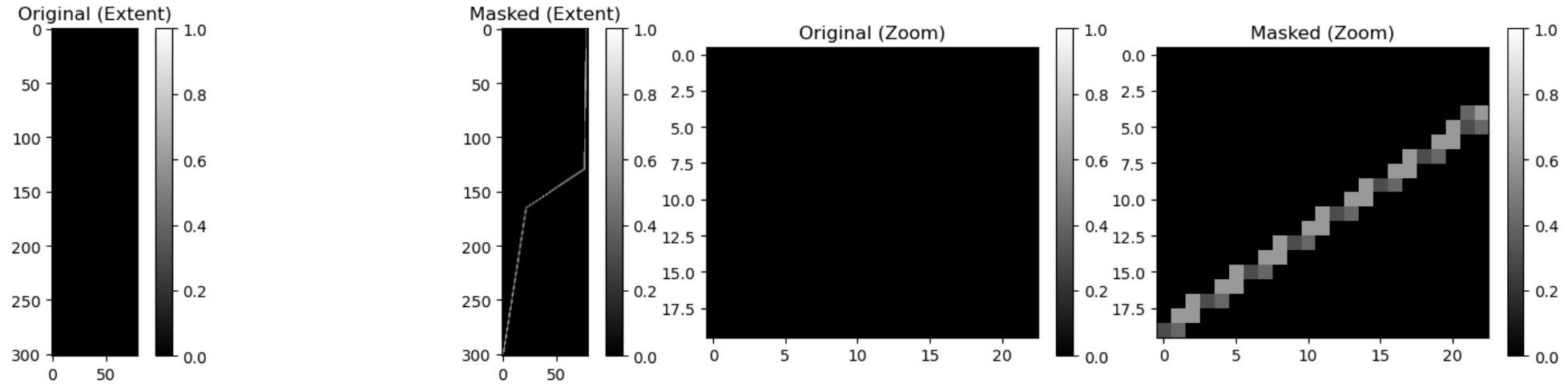
Scenario: New Highway in Milan

- **Objective:** Simulate the effect of a new highway on urban surface characteristics.
- **Inputs:**
 - ☐ Vector polygon for highway area (in .gpkg)
 - ☐ Rule: Apply mask to set land cover fractions (e.g., increase F_{AC} , reduce F_G)
- **Execution:**
 - ☐ `layer_alterator()` applied with `rules.json` and polygon attributes
 - ☐ Outputs generated: `F_AC_mask.tif`, `F_G_mask.tif`, etc.
- **Results:**
 - ☐ Clear spatial update of asphalt fraction (F_{AC})
 - ☐ Preservation of surrounding NoData
 - ☐ Ready-to-use outputs for further modeling (e.g., UHI impact)

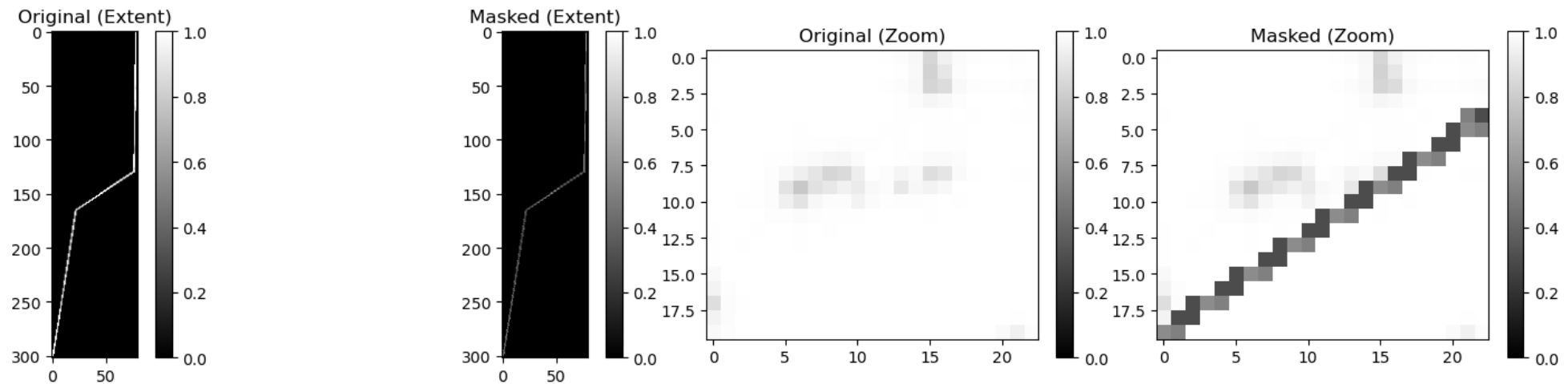


Demonstration

Comparison for BH.tif



Comparison for SVF.tif



Conclusions

Project Achievements

- Delivered a working tool for spatial raster updates via vector-defined rules
- Successfully supports both deterministic (mask) and relative (pct) operations
- Modular architecture with embedded validation ensures reliability
- Integrated into Jupyter for transparent, reproducible workflows

Impact & Future Work

- Valuable tool for urban planning, especially in Digital Twin contexts
- Enables “what-if” simulations with minimal overhead
- Future extensions:
 - ☐ GUI or API interface
 - ☐ Batch rule processing
 - ☐ Tiled processing for large datasets